# Final Year-End Project

***Masters of Data Science – Programming for Data Science Course***

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## Overview

This capstone project provides students with the opportunity to apply programming, data acquisition, and analytical skills developed throughout the course. Students will design and implement a **Streamlit web application** that processes and analyzes data from an **external API**, performs **exploratory data analysis (EDA)**, and integrates **Large Language Model (LLM)** capabilities through an API call.

The final deliverable should demonstrate teamwork, code collaboration, critical thinking, and practical application of modern data science tools.

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## Core Requirements

### 1. Data Acquisition

- Connect to at least one **external API** to fetch real data. This could be another source if available elsewhere.
- Clearly document the source and structure of the data (e.g., JSON, CSV, XML).
- Implement data retrieval and storage logic within the Streamlit app or a separate module.

**Examples:**

- Financial data (e.g., Alpha Vantage, Yahoo Finance)
- Weather data (e.g., OpenWeatherMap API)
- Social media data (e.g., Reddit)
- Public datasets (e.g., government open data APIs)

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### 2. Exploratory Data Analysis (EDA)

Perform an exploratory analysis that includes:

- Descriptive statistics and visualizations.
- Identification and treatment of outliers or missing data.
- Correlation analysis between key variables.

- Insights and summaries displayed within the Streamlit interface.

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### 3. Integration of a Large Language Model (LLM)

- The application must demonstrate interaction with a **Large Language Model (LLM)** API (open-source or closed-source).
- Options are:
    - Call LLM directly through API (OpenAI or Anthropic or Gemini)
    - Download local tiny LLM (see chapter 1 here)
    - Can utilize LLM either through Claude Code or Codex but explicitly say you're not using API and you want to use the available LLM that's part of the subscription.
- The LLM should be used meaningfully to **enhance insights or automate part of the analysis**, such as:
    - Summarizing data findings.
    - Generating natural-language reports.
    - Assisting with feature interpretation or user input queries.

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### 4. Streamlit Application

Your team must build a **Streamlit web app** to present your project. The app should:

- Have a clean, organized interface.
- Allow users to interact with data (e.g., filter, visualize, or query data).
- Display the results of your EDA.
- Integrate the LLM functionality directly within the app (e.g., chatbot, data summary, text generation).
- Include appropriate error handling, documentation, and user guidance.

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### 5. Code Quality and Collaboration

- Code must be modular, well-documented, and readable.
- Use of version control (Git/GitHub) is mandatory.
- Include a **README.md** file describing:
    - Project purpose and overview
    - Setup and installation instructions
    - API and data source details
    - How to run the Streamlit app

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## Deliverables

1. **GitHub Repository**
    - Contains all source code, data files (if applicable), and documentation.
    - Shows meaningful commit history from each team member.
2. **Streamlit App**
    - Hosted locally or via Streamlit Cloud.
    - Must demonstrate all required functionalities.
3. **Project Report (2–3 pages)**
    - Briefly describe project objectives, methodology, and results.
    - Include screenshots of the Streamlit app.
    - Reflect on team collaboration and lessons learned.
4. **Class live demo and presentation (max time 15 mins)**
    - Get in front of class to demo!

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# Evaluation Criteria

| Component | Weight |
|---|---|
| Data Acquisition and Exploratory Data Analysis (EDA) | 20% |
| Code Quality, Documentation, and GitHub Collaboration | 20% |
| LLM Integration and Usage | 20% |
| Streamlit App Functionality and Design | 20% |
| Presentation | 20% |

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# Submission

- Submit the **GitHub repository link** and **Streamlit app URL** (if hosted online).
- Include the **project report (PDF or Word)** in the repository and submission portal.